

Chapter 5

LRRK2 and Autophagy

Claudia Manzoni and Patrick A. Lewis

Abstract Leucine-rich repeat kinase 2 (LRRK2) has been implicated in a wide range of cellular processes, including the catabolic pathways collectively described as autophagy. In this chapter, the evidence linking LRRK2 to autophagy will be examined, along with how regulation of autophagy and lysosomal pathways may provide a nexus between the physiological function of this protein and the different diseases with which it has been associated. Data from cellular and animal models for LRRK2 function and dysfunction support a role in the regulation and control of autophagic pathways in the cell, although the extant results do not provide a clear indication as to whether LRRK2 is a positive or negative regulator of these pathways, and there are conflicting data as to the impact of mutations in LRRK2 causative for Parkinson's disease. Given that LRRK2 is a priority drug target for Parkinson's, the evidence suggesting that knockout or inhibition of LRRK2 can result in deregulation of autophagy may have important implications and is discussed in the context of our wider understanding of LRRK2.

Keywords LRRK2 • Macroautophagy • Catabolism • Parkinson's disease • Lysosomes • Autophagosomes

Introduction

LRRK2 has been linked to the cellular processes collectively called autophagy by a wide range of studies and in a number of model systems. While the precise mechanisms governing how LRRK2 interacts with or regulates autophagy remain obscure as well as the impact this may have in disease, it is indisputable that the manipulation of LRRK2 significantly alters the major pathways involved in autophagy. In this chapter, the evidence linking LRRK2 with autophagy is assessed, and the various potential scenarios as to how this might relate to disease are played out.

C. Manzoni • P.A. Lewis (✉)

School of Pharmacy, University of Reading, Whiteknights, Reading, UK

Department of Molecular Neuroscience, UCL Institute of Neurology,
Queen Square, London, UK

e-mail: c.manzoni@reading.ac.uk; p.a.lewis@reading.ac.uk

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Autophagy

To understand the role of LRRK2 in autophagy, it is first essential to briefly summarize what is meant by this term. Autophagy (from the Greek *auto*—self—and *phagein*/phagos, eat) is an essential catabolic process for the degradation of exhausted, damaged, and potentially dangerous materials within eukaryotic cells [1]. Since materials to be degraded range from misfolded proteins to damaged organelles, autophagy can be seen as a routine process of cleaning and waste removal. Specialized cells, such as immune cells able to perform phagocytosis, specifically use autophagy to destroy the potentially hazardous content of the phagocytic vesicle; therefore, in this respect, autophagy can also be seen as a first-line defense for keeping a clean and safe cellular environment [2]. Another key connection is between autophagy and cell death; a type of programmed cell death (PCD) called PCD type II occurs by hyperactivation of autophagy, where lysosomes and vacuoles take over and degrade the whole cellular body [3]. Finally, autophagy can be initiated under a cocktail of stress and nutrient depletion; under these circumstances, the self-digestion of portions of cytoplasm becomes a fundamental buffer system for recycling amino acids and fuelling cell energy with substrates for oxidation [4]. Modified cells in which autophagy has been ablated do not survive starvation periods, emphasizing the essential nature of this function for the evolution of life.

It is quite clear from the above that autophagy is a fundamental, and complex, process—and the molecular mechanisms behind it are equally complex. A wide variety of signals, checkpoints, molecular players, and feedback loops are involved in its regulation, and our understanding of these is far from complete [5]. Moreover, even if the end point of autophagy is the delivery and degradation of materials into the lysosomes, several routes exist to achieving that end point: macroautophagy, chaperone-mediated autophagy (CMA), and microautophagy (Fig. 5.1).

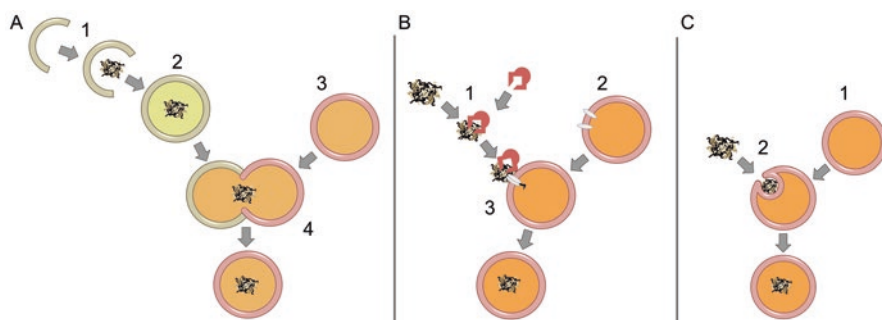


Fig. 5.1 The different forms of autophagy. (a) Simplified scheme for macroautophagy: (1) formation of omegasome, (2) encapsulation of waste for degradation, (3) lysosome, (4) fusion of autophagosome with lysosome. (b) Simplified scheme for chaperone-mediated autophagy: (1) HSC70 recognizes and binds to waste for targeting, (2) Lamp2A forms complex in membrane of lysosomal vesicle, (3) protein for degradation is passed into the lumen of the lysosome via Lamp2A complex. (c) Simplified scheme for microautophagy: (1) lysosome, (2) invagination of lysosomal membrane allows direct sequestration of waste for degradation into the lumen of the lysosome

Macroautophagy is probably the best-known, and certainly most studied, form of autophagy; it is characterized by the production of a specific double-membraned vesicle called an autophagosome. The vesicle production starts in discrete parts of the endoplasmic reticulum called omegasomes. The cup-shaped membrane that emerges from the omegasome is referred to as isolation membrane or phagophore; it will eventually detach and close around the material to be degraded thus sequestering portions of the cytosol and, in some cases, organelles. Non-specific, bulk macroautophagy occurs when parts of the cytosols are non-specifically engulfed during the closure of the autophagosome. Specific macroautophagy occurs when the autophagosome chooses the material to be degraded; in this case cargo proteins (such as p62, NBR1, MDP52, optineurin, and NIX) recognize the material to be degraded, tag this material, and simultaneously interact with microtubule-associated proteins 1A/1B light chain 3A (more commonly referred to as LC3), a protein which is specifically inserted into the nascent vesicle. The generation of this bridge is what eventually guides the autophagosome to close around the cargo to be degraded. The fully loaded autophagosome is then dispatched for degradation following a final fusion with the lysosome. In the process referred to as *canonical macroautophagy*, the phosphorylation state of the UNC-like kinase 1 (ULK1) complex drives the creation of the isolation membrane [6]. The ULK1 complex is directly phosphorylated by the mammalian target of rapamycin complex 1 (mTORc1) and AMP kinase (AMPK). mTOR is sensitive to nitrogen availability, and amino acid depletion is able to activate autophagy through mTOR, while AMPK is sensitive to cellular energy levels and is activated by depletion of ATP, along with the concomitant increase in ADP and AMP. The second key player in canonical macroautophagy is the BECLIN-1 complex, able to activate the class III phosphatidylinositol 3-kinase (PI3K) VPS34 necessary for the production of inositol-3 phosphate (I3P) that is enriched at the omegasome, remains in the isolation membrane, and promotes the recruitment of effector proteins, such as the WIPI proteins, to the autophagosome [7]. Finally, the third crucial player in canonical macroautophagy is the aforementioned LC3, a protein that undergoes proteolytic cleavage and conjugation with phosphatidylethanolamine prior insertion into the autophagosome membrane [8]. LC3 is important for specific cargo recognition and for the final closure of the autophagosome. Moreover, since its modifications are detectable as a shift in electrophoretic mobility, LC3 has become the marker of choice for biochemically monitoring autophagosome formation [9].

The description of canonical macroautophagy suggests that *noncanonical* pathways also exist, and indeed this is the case. Examples of noncanonical pathways are those that do not require ULK1 (e.g., macroautophagy induced by ammonia), BECLIN-1 (macroautophagy induced by MK801 and arsenic trioxide), or even LC3 -since small monomeric GTPases like RAB9 may sometimes take its place [10]. Noncanonical macroautophagy is not well understood, and we still do not have a detailed comprehension of the stimuli able to trigger these alternative pathways nor of the protein complexes involved in their regulation.

In contrast to macroautophagy, *CMA* does not require the generation of autophagic vesicles: the materials to be degraded, for the most part proteins, are delivered directly to the lysosomes. Proteins containing the pentapeptide motif KFERQ are recognized by the heat shock cognate protein 70 (HSC-70) and transported to the lysosomal surface where this chaperone is recognized by the lysosomal protein LAMP2A and the cargo internalized to be degraded [11]. The process of CMA has been investigated in a great deal of detail with regard to disorders such as Alzheimer's, the prion disorders, and Huntington's chorea, where deposition of amyloid material is seen as a consequence of an imbalance between production and catabolism of amyloidogenic proteins that are known substrates of CMA. This is likewise the case for proteins involved in Parkinson's, for example, α -synuclein and LRRK2 (of which more later).

Finally, *microautophagy* occurs through direct invagination of the lysosomal membrane, with target proteins/waste imported straight into the lysosome. Little is known about microautophagy [12]. It is likely that it represents a constitutive form of autophagy; however, starvation and inhibition of mTORc1 by rapamycin are able to potentiate it. One explanation is that microautophagy acts to balance the dimensions of the lysosome, which would otherwise continuously increase its surface by fusion with autophagosomes and late endosomes.

To add another layer of complexity, autophagy can also be classified by the cargo targeted for degradation, for example, *mitophagy* for mitochondria, *pexophagy* for the peroxisomes, *xenophagy* for the degradation of engulfed, exogenous foreign bodies in phagocytic cells, and so on [13–15].

Evidence for a Role for LRRK2 in Autophagy

Introductory Note

There are a wide range of data supporting a role for LRRK2 in the regulation and function of autophagy. At this juncture, and before proceeding any further, it is worth considering several aspects of exactly how LRRK2, a large and complex protein, interacts with autophagy, a likewise complex and dynamic process [16].

First, the distinction between the physiological function of LRRK2 and its role in disease needs to be considered. The evidence suggesting the involvement of LRRK2 in autophagy, while not conclusive, is extensive; however, how this involvement relates with disease is still a matter of great discussion. We face three possible scenarios:

1. The role of LRRK2 in autophagy may be relevant in physiology, but it may not be related with the molecular mechanism of disease.
2. Alterations of autophagy due to changes in LRRK2 may influence the disease, but this may not be the primary event leading to disease.
3. Finally the alteration of autophagy by LRRK2 may truly be the primary, causal, molecular mechanism of disease.

It is unclear which scenario holds true, and the situation is complicated by the involvement of LRRK2 in diseases other than Parkinson's (cancer, inflammatory bowel disorder, and leprosy), where the contribution of autophagy to pathogenesis may be distinct from that in PD [17].

A second remark relates to the experimental approaches used to assess LRRK2 and autophagy. Experiments are conducted in many different model systems; data in the literature are mixed—comparing results from endogenous LRRK2, overexpression models, and ablation of kinase, GTPase, or both activities alongside the analysis of Mendelian mutations that are relevant to Parkinson's disease. The heterogeneous experimental system, in combination with the nuances of the autophagic process itself, accounts for the disturbing amount of conflicting results in the literature. This does not, per se, invalidate the extant literature; however, it represents a major challenge in terms of identifying a unifying role for LRRK2 in autophagy and acts as a cautionary note with regard to extrapolating conclusions from single model systems and unique experimental settings.

The Case of Parkinson's Disease

Alterations of Autophagy in Parkinson's Disease Brains

Parkinson's disease (PD) is pathologically defined by the deposition of misfolded protein, predominantly α -synuclein, in the brain as intracellular inclusions known as *Lewy bodies* [18]. The accumulation of undegraded proteins and their deposition in amyloid bodies are common to a number of different misfolding disorders frequently referred to as amyloidoses. This phenomenon has been seen as consequence of an intrinsic imbalance between protein production and turnover. Since degradation of misfolded proteins is carried out both by the proteasome and the autophagic systems, the presence of amyloid deposits like Lewy bodies in PD can be taken as indirect evidence of alterations in proteostasis and, by implication, autophagy. The majority of PD cases are idiopathic, also called sporadic, since they happen with no proof of causality for Mendelian transmission. Accumulation of autophagic vesicle reminiscent of a possible impairment of macroautophagy, as well as reduction in the LAMP2A and HSC-70 protein levels important for CMA, has indeed been described in postmortem human specimens from idiopathic PD cases [19]. Moreover, autophagy-deficient mouse models display brain abnormalities like formation of ubiquitin-positive inclusions and presynaptic accumulation of α -synuclein and LRRK2, further emphasizing autophagy as one of the cellular mechanisms contributing to the general pathogenesis of PD [20]. A small percentage (between 5 and 10 %) of PD cases are familial, caused by a mutation transmitted through generations in a Mendelian fashion. Of the Mendelian causes of Parkinson's, LRRK2 is the most frequently mutated gene [21]. The case of familial PD associated with LRRK2 mutations is quite unusual due to the variability in protein deposition observed in postmortem brains [22]. The majority of LRRK2 (PD) cases present with deposition of α -synuclein as observed in idiopathic PD cases; however, a small minority of

patients show inclusions of proteins other than α -synuclein—such as the microtubule-associated protein tau or the RNA-binding protein TDP-43—or no identifiable inclusions at all [23]. Whether this pleomorphic pathology is indicative of a fundamental disruption of aggregate disposal mechanisms, including autophagy, due to LRRK2 mutations is unclear.

Alterations of Autophagy in LRRK2 Mutant PD Cell Models

A number of different cellular approaches have been used to examine autophagy in the context of LRRK2 (PD) mutations using multiple cellular systems. The first studies investigating LRRK2 and autophagy did so using immortalized cell lines expressing either exogenous wild type or mutant LRRK2. Plowey and co-workers, using SHSY-5Y neuroblastoma cells as a model, discovered alterations in autophagic vesicles upon transfection with LRRK2 containing the G2019S pathogenic mutation [24]. Examining a previously reported LRRK2 phenotype, neurite retraction, the authors were able to present alterations in this by knocking down key components of the autophagy machinery such as LC3 or ATG7—providing the first direct evidence of a link between LRRK2 and macroautophagy. Soon after, a group based at the University of Oxford demonstrated that LRRK2 regulates autophagic vesicle formation and associates with specific vesicle microdomains [25]. Following these early reports, data has accumulated implicating LRRK2 in the regulation of macroautophagy; however, there is as yet no consensus on the molecular mechanism underlying this association with some papers suggesting that this could be through regulation of calcium levels and the NAADP pathway [26], the MEK/ERK pathway [27], and via the Beclin-associated protein BCL-2 [28]. As tool compounds targeting LRRK2 became available, enabling researchers to effectively inhibit LRRK2 kinase activity, it also became clear that reducing LRRK2 activity in this manner acted to induce autophagy [29, 30]. Likewise, knockdown of LRRK2 has been reported to cause an induction of autophagy [25]—although in both cases (knockdown and inhibition) the precise mechanism remains obscure. There appears to be a close relationship between LRRK2 kinase activity and cellular localization, and a study utilizing BV2 and RAW cells (microglial and macrophage in origin, respectively) highlighted an intimate connection between membrane localization of LRRK2, kinase activity, and regulation of macroautophagy [31]. Intriguingly, given the complex enzymatic complex formed by LRRK2, a yeast-based study revealed a role for the GTPase domain of LRRK2 in regulating autophagy [32].

One approach to directly test how mutations in LRRK2 impact on autophagy in disease context is to use cells derived from PD patients, compared with appropriate nonmutant controls. *Fibroblasts* have been used as a primary experimental model system as well as a source for *induced pluripotent stem (iPS)* cells to be then differentiated into human neuronal populations. Data generated through the use of these models are highly variable and discordant and (similar to the models described above) fail to identify a clear molecular mechanism for LRRK2 in disease. Taken in

toto, however, they present again an undeniable association between LRRK2 and autophagy. Fibroblasts carrying the G2019S mutation in the kinase domain of LRRK2 have been shown to have an increased basal rate of macroautophagy. This alteration of the basal flux was not related with the canonical control of mTOR over macroautophagy, but was dependent upon an abnormal activity of the MEK1/3 pathway [27]. Another mTOR-independent alteration of macroautophagy was reported in human fibroblasts with PD mutations encompassing the entire catalytic core of the LRRK2 protein—suggesting that at an endogenous level, diverse LRRK2 mutations could alter autophagic response [33]. Although this latter study did not observe a basal increase in macroautophagy as previously seen in G2019S carriers, impaired macroautophagic response to starvation was detected for all mutant lines. G2019S mutant fibroblasts have also been documented to have a raised sensitivity to inhibition of mitochondrial fission, a sensitivity that is linked to autophagy [34], and also to the mitochondrial toxin 1-methyl-4-phenylpyridinium (MPTP)—like-wise dependent upon autophagy [35].

An early study presenting data from neurons derived from LRRK2 mutant patient iPS cells suggested that the G2019S mutation could modify autophagic vesicle formation in dopaminergic neuronal cells [36], although this was matched by data from idiopathic PD patient-specific cells (i.e., not carrying a genetic defect), making interpretation of these results complicated.

Using patient-derived neuronal cells, LRRK2 has also been implicated in CMA [37]. In this model system, LRRK2 was degraded via CMA, with proof of mutant LRRK2 acting to inhibit this pathway. Given the evidence linking α -synuclein to CMA for its degradation, this provides a potential mechanism for protein aggregate buildup in disease—although it should be noted that LRRK2 has been reported to be degraded by the ubiquitin proteasome system as well [38], and how these pathways intersect and compensate each other in the context of LRRK2 is still not clear.

Alterations of Autophagy in LRRK2 PD Animal Models

In vivo models for genetic forms of PD have, almost without exception, proved disappointing in terms of reproducing disease-relevant phenotypes—most notably clear nigral cell death and protein aggregation. Despite this, there are large numbers of animal models for LRRK2 disease ranging from *Caenorhabditis elegans* to rodent transgenics. Data from these models have proved an important source of evidence for a link between LRRK2 and autophagy, although again the precise molecular events linking the two are ambiguous. *Caenorhabditis elegans*, worms expressing mutant LRRK2 (both the R1441C and G2019S mutations), displayed reduced autophagic activity, while wild-type LRRK2 improved autophagic function throughout aging [39]. In *Drosophila melanogaster*, knockout of the *Drosophila* LRRK2 paralog (*dLRRK*) caused severe deficits in the autophagy/lysosomal system—with accumulation of enlarged, dysfunctional lysosomes [40]. A caveat to both worm and fly studies is that neither *C. elegans* nor *D. melanogaster* possesses clear

LRRK2 paralogs—each possessing only one LRRK gene. Whether these species provide a suitable model for LRRK2, or LRRK1 for that matter, is not yet clear [41].

The generation of *LRRK2 rodent knockout* models appeared at first to be an underwhelming avenue of investigation: the absence of neurodegeneration and the complete lack of pathological hallmarks typical of PD in the brain of LRRK2 knockout animals led to these models being dismissed as something of a footnote to the LRRK2 field. Once investigators looked beyond the brain, however, intriguing data began to emerge. Two independently generated LRRK2 knockout mice from the Shen laboratory displayed peculiar alterations in their kidneys, with signs of enhanced autophagy at young ages and reduced autophagy at 20 months accompanied by progressive accumulation of secondary lysosomes, α -synuclein, and ubiquitinated proteins [42]. A third LRRK2 knockout mouse was generated at the Novartis laboratories [43]. A very similar alteration in the kidneys was reported, with accumulation of secondary lysosomes even though the absence of LC3 alterations, the presence of mTOR changes, and the end point of the study at only 14 months make it difficult to fully understand the implication for macroautophagy. Interestingly, changes were detected in the lungs where lamellar bodies accumulated abnormally in type II cells. A fourth LRRK2 knockout mouse reproduced a similar profile of progressive kidney alterations. Interestingly, accumulation of the autophagy marker p62 was similar to that observed in the Shen models, while LC3-II was shown to accumulate rather than decrease as previously reported [44]. Similar to the data from mice, age-dependent results with accumulation of lysosomes in the kidneys and lamellar bodies in the lungs have been subsequently reproduced by at least two independent groups in a LRRK2 knockout rat generated by the Sigma Advanced Genetic Engineering (SAGE) laboratories [45]. Although the comparison between knockout of LRRK2 and inhibition of its kinase activity is not a simple one (noting that LRRK2 has a number of other activities that are lost if the gene is knocked out), it is striking that treatment of nonhuman primates with a specific inhibitor of LRRK2 kinase activity recapitulates many of the characteristics of the knockout phenotype [46]. In particular, the acutely treated primates developed vacuolar pathology in their lungs, with potential consequences for lung function and striking a note of caution with regard to the use of LRRK2 kinase inhibitors in a clinical setting.

There exist a range of rodent LRRK2 mutation models, using a variety of strategies (knock in and overexpression) to assess the impact of LRRK2 mutations. A number of these have reported alterations in autophagic markers such as LC3 and p62, although these appear not to be consistent between mutations studied. For example, G2019S mice exhibit alterations in these markers [47], whereas mice expressing R1441C do not [48]. As for the cellular studies of LRRK2 and autophagy, the conclusion to be drawn from these animal models is that there is proof of an impact of LRRK2 on autophagy, but that it is not obvious from the extant experimental data as to the precise mechanism or direction of this alteration.

Finally, and providing a valuable illustration of the reciprocal nature of the pathways that govern autophagy, selective knockout of *Atg7*, a key component of the macroautophagy machinery, in dopaminergic neurons results in the accumulation of LRRK2 and α -synuclein [20]. This study demonstrates that, just as LRRK2 can alter autophagy, autophagy can have an impact on LRRK2.

Parkinson's Disease: A Possible Scenario

As with much of the science relating to LRRK2, it is important to examine the biological insights in the context of the preeminent driver of research into this protein: Parkinson's disease. The vast majority of PD cases are idiopathic in nature, with no positive familial history of disease and with no clear association with genetic risk factors. The presence of Lewy bodies, predominantly composed of undigested, misfolded, and ubiquitinated α -synuclein, paired with the finding of vacuolation and alterations in CMA in idiopathic brains, provided a clear indication that autophagy could be implicated in the etiology of PD. However, the absence of specific genes to investigate built a nearly insurmountable wall between the hypothesis and the actual possibility of proving it with further investigations.

A small percentage of PD cases, however, are familial in origin and are therefore associated with mutations in known genes. LRRK2 is the gene product most frequently mutated in familial PD but other mutated proteins have been identified: α -synuclein, GBA, VPS35, PINK1, Parkin, DJ-1, and FBXO-7 to name just a selection [21]. Unlike the idiopathic cases, familial cases are easier to study in model systems, and the autophagy theory can be dissected in greater depth.

In this chapter we have focused on how LRRK2 has been linked to autophagy; it would therefore be interesting to consider how LRRK2 and autophagy relates to other PD proteins. The most prominent of these is α -synuclein, with mutant α -synuclein having been demonstrated to impair CMA in a process that highly resembles the one proposed as justification for LRRK2 inhibition of CMA, and equally aggregated α -synuclein can inhibit autophagy [49, 50]. GBA is a lysosomal enzyme, important for lysosomal integrity and functionality. When altered, the generation of a primary lysosomal defect leads to impaired autophagy and mitophagy [51]. The same can be said for ATP13A2, an ATPase involved in cation transport in acidic vesicles, associated with Kufor Rakeb syndrome, a form of dementia that presents with parkinsonism [52]. Loss of function mutations in ATP13A2 result in a severe lysosomal deficiency and a reduction in autophagosomes clearance. VPS35 is a well-known component of the retromer complex that is implicated in the trafficking of proteins and enzymes in the trans-Golgi network thus supporting the correct loading of lysosomes with hydrolytic enzymes [53]. PINK1, Parkin, and FBXO-7 have all been implicated in mitophagy, and it is now well established they have a fundamental role in mitochondria quality control [14]. Finally WDR45 is a protein associated with a form of neurodegeneration with brain iron accumulation that presents with parkinsonism [54]. WDR45 has not been extensively studied, but it is often referred to as WIPI-4, and we know at least other two proteins, WIPI-1 and WIPI-2, that are important for the initial stages of macroautophagy suggesting a strong link to autophagy.

By looking at a broad spectrum of familial PD and parkinsonism, it seems like there is a common theme associated with the disease. All familial genes are for some extent associated with catabolism [55, 56]. Interestingly this association is not sharp or precise; it is not a single degradation pathway or a single form of autophagy

that is being implicated. The leitmotif behind different forms of familial PD is much more broad, but it points in the direction of impairment of catabolic processes related with autophagy, lysosomes, and possibly vesicle recycling. This may account for the clinical and phenotypic differences observed with different mutations; it may also suggest that rather of being a strong molecular mechanism of disease, alterations in catabolism may act by creating a susceptible environment in which a combination of other genetic and environmental factors may push the system above the threshold eventually culminating in the propagation of neurodegeneration.

Further Diseases, Proteins, and Circumstantial Evidence

One of the most fascinating aspects of the LRRK2 gene and human pathobiology is the implication of LRRK2 in multiple human disorders. While the bulk of research into LRRK2 has focused on its role in PD, it has also been linked to inflammatory bowel disease, cancer, and leprosy [17]. It is notable that autophagy is a unifying feature of all of these disorders. Unfortunately there are very few papers investigating the molecular connection of LRRK2 with the pathogenesis of these diseases leaving only genetic evidence, most certainly circumstantial in nature, to suggest that LRRK2 may play a role in the regulation of autophagy and that this may consequently impact on the pathogenesis of disease. An isolated publication reported a possible molecular association of LRRK2 and the receptor tyrosine kinase MET signaling in cancer, thus implicating LRRK2 with alterations in apoptosis and autophagy [57].

It is also noteworthy that at least one other protein in the ROCO family, to which LRRK2 belongs, has been strongly implicated in the regulation of autophagy. Death-associated protein kinase 1 (DAPK1) has been closely linked to type II PCD and has been reported to directly regulate the BECLIN-1 pathway by phosphorylation of the BECLIN-1 protein complex [58]. How this relates to the changes observed in LRRK2 models, however, has not been investigated.

Conclusions

Mechanisms Linking LRRK2 to Autophagy

How do we bring together these discordant data to gain insight into the role of LRRK2 as an orchestrator, or at least conductor, of autophagic activity in the cell? The extant data relating to LRRK2 and autophagy are summarized in Table 5.1. This highlights the volume of evidence linking LRRK2 to autophagy while also emphasizing the challenge presented by attempting to dissect out the precise mechanisms and pathways involved.

Table 5.1 Alterations in autophagy pathways linked to LRRK2

Autophagy type	Model	Impact	Citation
Macroautophagy	SHSY-5Y cells, mutations (G2019S, K1906M)	Increased autophagosomes in presence	Plowey et al. [24]
Macroautophagy	HEK293T cells, VERO cells, mutations (R1441C/G2019S), knockdown	Impairment of autophagosome formation due to R1441C mutation, increase in autophagic activity due to LRRK2 knockdown	Alegre-Abarrategui [25]
Macroautophagy	HEK293T cells, PC12 cells, mutations (G2019S, K1906M)	Increased autophagosome formation	Gomez-Suaga et al. [26]
CMA	SHSY-5Y cells, HEK293 cells, human-derived iPSC neuronal cells, mutations (G2019S)	Inhibition of CMA by mutated LRRK2	Orenstein et al. [37]
Macroautophagy	Human fibroblasts, mutations (G2019S)	Increase in basal autophagy	Bravo-San Pedro et al. [27]
Macroautophagy	HeLa cells, human fibroblast, mutation (G2019S)	Increased autophagy due to mutation, Bcl-2 mediated	Su et al. [28]
Macroautophagy	H4 neuroglioma cells, primary rat astrocytes, LRRK2 kinase inhibition	Increase in autophagic flux	Manzoni et al. [29]
Macroautophagy	RAW264.7 cells, BV2 cells, LRRK2 knockdown, LRRK2 kinase inhibition	Decrease in rapamycin-induced autophagic flux	Schapansky et al. [31]
Macroautophagy	Human-derived iPSC neuronal cells, mutations (G2019S)	Accumulation of autophagic vesicles, decreased lysosomal function	Sanchez-Danes et al. [36]
Macroautophagy	<i>C. elegans</i> , mutations (G2019S)	Accelerated loss of autophagic function due to G2019S mutation	Saha et al. [39]
Macroautophagy	<i>Mus musculus</i> , mutations (G2019S transgene)	Enlarged autophagic vesicle-like structures	Ramonet et al. [47]
Macroautophagy	<i>Saccharomyces cerevisiae</i> , mutations (R1441C, artificial GTPase mutations)	Increased autophagic vesicles due to loss of GTPase activity	Xiong et al. [32]

(continued)

Table 5.1 (continued)

Autophagy type	Model	Impact	Citation
Macroautophagy	<i>Mus musculus</i> , knockout, knock in kinase dead, mutation (G2019S)	Autophagic/lysosomal abnormalities due to loss of LRRK2	Herzig et al. [43]
Macroautophagy	<i>Drosophila melanogaster</i> , knockout	Accumulation of autophagosomes due to loss of <i>dLRRK</i>	Dodson et al. [40]
Macroautophagy	SHSY-5Y cells, LRRK2 kinase inhibition	Induction of autophagy following inhibition of LRRK2 kinase activity	Saez-Atienzar et al. [30]
Macroautophagy	Human fibroblasts, LRRK2 mutations (R1441G, Y1699C, G2019S)	Reduced autophagic response to starvation due to mutations.	Manzoni et al. [33]
Macroautophagy	<i>Mus musculus</i> , knockout	Altered autophagic markers, vesicle accumulation	Tong et al. [42]
Macroautophagy	<i>Mus musculus</i> , knockout	Altered autophagic markers, vesicle accumulation	Hinkle et al. [44]
Macroautophagy	<i>Macaca fascicularis</i> , LRRK2 kinase inhibitors	Altered lysosomal function	Fuji et al. [46]
Macroautophagy	<i>Rattus norvegicus</i> (Long-Evans rat), knockout	Altered lysosomal function	Baptista et al. [45]

As a foundation to our understanding of LRRK2 in this context, it is worth revisiting the presumed role of LRRK2 within the cell. Based upon the functional domains that make up LRRK2, i.e., protein-protein interaction/scaffolding domains, a GTPase domain, and a kinase domain, the working hypothesis for LRRK2 function is that it operates as a signaling node within the cell, controlling cellular processes via phosphorylation of substrates in signaling pathways or by acting as a molecular switch for other kinases via the ROC/GTPase domain. As noted above, the complex domain organization of LRRK2 makes it difficult to be categorical about which of these activities predominates or whether there is a sequential series of events within LRRK2 that leads to stimulation/suppression of downstream pathways.

It is clear that there are alterations in autophagic function that can be measured as a consequence of changes in LRRK2; however, no specific event has been identified close to the known regulators of autophagy. This, in turn, raises an important question as to whether the changes that are observed in autophagy are proximal, immediate events to LRRK2 or whether LRRK2 is acting distal to these changes, perhaps a number of signaling events removed. The second key question is what

aspects of LRRK2 biology are critical for these cellular phenotypes—is kinase activity the central player? Or GTPase switch activity? With the available published data, it is impossible to rule out any of these mechanisms. Indeed, there is potential for LRRK2 to act simultaneously as a regulator of the initiator of autophagic processes, perhaps by a kinase-mediated signaling cascade, while also influencing CMA and autophagosome recycling through a Rab-like GTPase function. Finally the heterogeneity of data obtained in different experimental systems may stimulate the suspicion that LRRK2 may have different roles in different cell types; thus, the relation of LRRK2 and autophagy could be cell/tissue and maybe time dependent rather than a simple general feature.

It is certainly the case that further research is justified and perhaps even required to clarify which of these molecular scenarios is correct.

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